

DUSTRUNNER · REQUIREMENTS

DustRunner requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/dustrunner		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.6	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.5	2026-09-30	Amish Chadha	Status from DRN-CAL-001 v0.3 for the constructable design (DRN-DDR-003); R10 (mass), R13 and R14 not met. No target changed
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002). R9 restated with a 6 m/s start limit and 8 m/s abort limit; R10 allows 75 N for the row-entry transient; status from DRN-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	Apply DRN-DDR-001 (R14 restated for rows of 60 m or more; R13 covers the anemometer; reference row and table format kept); status from DRN-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with a reference row and status against the concept
0.1	2026-09-24	Amish Chadha	Initial scaffold

DustRunner requirements

These requirements are checked by calculation in DRN-CAL-001 v0.3, for the constructable design of DRN-DDR-003. On paper, five are met, three are at risk, three fall short of their targets and four cannot be verified at TRL 3. The three that fall short follow from the parts added to make the design buildable: R10 (the robot weighs 15.9 kg against 15 kg, though every wheel load is within its limit), R13 (estimated parts cost \$573, USD 73 over the value-engineering target of \$500) and R14 (3.2 years on a 60 m row; met from 63.5 m). No target has been changed; whether to restate R10 and R14 is open, and the cost savings worth trying are in the value engineering section of the design decisions register (DRN-DEC-001). The three at risk are R4 (5 mm frame steps and frame flange width), R9 (traction in wind along the row, margin 1.30 at the 6 m/s start limit but 0.98 if friction is only 0.3) and R11 (pack temperature while charging in the sun). R2, R3, R6 and R15 need tests or trials. Targets are still proposals for review, not validated with users or a pilot site, except where DRN-DDR-001 or DRN-DDR-002 records Amish's decision.

The **reference row** used throughout, kept by DRN-DDR-001 (D7), is one fixed-tilt table, one module high in portrait (1P), with 35 modules of 2,278 x 1,134 mm and about 580 W each: about 40 m long, about 90 m² of glass and about 20 kWp, tilted 25°, with the lower glass edge about 0.6 m above the ground. The first supported table format is 1P portrait with 2,278 mm modules (DRN-DDR-001, D10).

Table 1. Requirements and status at TRL 3.

ID	Requirement	Target	Status (DRN-CAL-001)	Verification (TRL 3 or later)
R1	Water-free cleaning	No water or liquid used in normal operation	Met by design	Design review
R2	Cleaning effectiveness on dry dust	Daily dry pass keeps the average soiling loss at 1.5 % or less when the site soiling rate is 0.3 to 0.5 % per day	Not verifiable at TRL 3; depends on dust type and dew. R14 on the 60 m row would need a residual below 1.3 %	Later soiling-station comparison
R3	Glass and coating safety	Only microfiber touches the glass; brush interference 3 to 5 mm along the whole brush; no hard bristles, grit traps or metal within 10 mm of the glass; complies with the module maker's cleaning guidance	Not verifiable at TRL 3; geometry met (3.5 to 4.4 mm with a 50 mm core; nothing else within 10 mm); abrasion over years unknown	Maker guidance review; later abrasion test on sample glass
R4	Fit the reference table	1P portrait modules 2,278 mm long with 30 to 35 mm frames; short (top and bottom) edges free of clamps; tilt 10 to 35°; gaps between modules up to 25 mm; frame height steps up to 5 mm	At risk: 25 mm gaps pass; a 5 mm step needs about 9 N against about 12 N spare traction; the wheel tread needs a visible frame flange of about 21 mm or more	Survey of target tables and frame profiles
R5	Coverage	95 % or more of the glass area brushed per pass	Met: 98.4 % on the 40 m row	Model check
R6	Rail-free installation	No rails along the row; only a dock at one end and clamp-on end stops at the other; no drilling of modules or structure; installed by two people in 60 min or less	Not verifiable at TRL 3; rail-free and clamp-on by design; time unverified	Installation trial
R7	Cycle time	A 100 m row cleaned out and back in 20 min or less	Met: 17.4 min	Calculation
R8	Energy autonomy	Three daily cycles on a 100 m row with no sun; recharged from the dock panel in one day at 3 peak sun hours	Met: 68 of 102 Wh usable; 45 Wh harvested against 23 Wh a day	Energy calculation
R9	Stay on the row	Two independent means stop the robot at each row end; hook rollers keep it on the table in operating wind; a run starts only when the dock anemometer reads below 6 m/s and aborts to the dock above 8 m/s (DRN-DDR-002, N1); robot parked and latched in the dock survives 35 m/s	At risk: parked, latch and end stops hold with factors over 4; traction margin 1.30 at the 6 m/s start limit and 1.07 at the 8 m/s abort limit at friction 0.4, but 0.98 and 0.81 at friction 0.3	Wind load calculation; later friction measurement
R10	Load on the modules	Robot mass 15 kg or less; each wheel load 60 N or less in steady running and 75 N or less for the brief transient as the lead wheels enter the first module (DRN-DDR-002, N2), subject to the module maker's frame loading guidance; carried on the frame only; no load on the glass except brush contact	Not met on mass: 15.9 kg with the parts added for construction. Wheel loads met: 59 N per wheel brushing at worst tilt with 60 N hook preload; about 73 N for about a second at row entry	Mass estimate; module maker's guidance

ID	Requirement	Target	Status (DRN-CAL-001)	Verification (TRL 3 or later)
R11	Environment	Electronics IP65; operates at 0 to 50 °C ambient on glass up to 75 °C; UV-stable plastics; dust-sealed bearings	At risk: the parked pack may exceed its 45 °C charging limit in the sun; a sunshade is added	Datasheets and design review; later thermal check
R12	Safe to be near	Emergency stop button on each end truck; brush and drives stop within 2 s of a stop command or of a truck lifting off the frame; brush ends guarded by the hood; nothing on the robot can reach PV cables, connectors or junction boxes	Met on paper: brush stops in 0.13 s, robot in 0.07 s; belt guards and a stop button in each truck plate	Design review; later bench test
R13	Affordable	Robot, dock and anemometer for one row, parts at or below the value-engineering target of \$500 (<code>budget_usd</code> , a hypothetical control target)	Over the value-engineering target by USD 73: \$573.00 indicative after the parts added for construction (DRN-DDR-003); the target stays at \$500 (DRN-DDR-002, N3); savings to try are in DRN-DEC-001	Priced BOM (<code>bom/bom.csv</code>)
R14	Pays for itself	Value of the recovered energy covers the parts cost in 3 years or less on rows of 60 m or more, mid case (0.3 % per day soiling, monthly manual wash as baseline, \$0.10/kWh)	Not met: 3.2 years on a 60 m row at \$573; met on rows of 63.5 m or more; 1.9 years on a 100 m row (4.7 years on the 40 m reference row, which is below the scope of R14)	Energy yield calculation
R15	Serviceable	Microfiber sleeve replaced in 15 min or less without taking the robot off the row; motors, bearings, wheels and battery are standard catalog parts	Not verifiable at TRL 3; catalog parts by design	Later service trial

■ Assumptions

- Specific yield about 5.5 kWh per kWp per day for a clean array at a sunny desert site (estimate).
- Soiling rate 0.3 % per day (mid case) and 0.49 % per day (high case, as measured at QEERI in Qatar; see DRN-PRB-001).
- Baseline practice is a manual wash every 30 days. Longer manual intervals, or no cleaning at all, make DustRunner's case stronger.
- Electricity or displaced diesel valued at \$0.10/kWh. Diesel-displacing mini-grids are often higher.
- 20 W dock panel, 75 % derating for heat, dust, wiring and a simple charge controller.
- Friction coefficients, brush pile pressure and the visible frame flange width are assumptions listed in DRN-CAL-001, Table 1, and must be measured.